AUTOMATIC ESSAY EVALUATION

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Problem Statement

Manual essay evaluation is a resource-intensive and error-prone process, subject to human bias and inconsistencies. As the volume of written content increases across educational institutions, businesses, and various professional fields, traditional evaluation methods struggle to scale effectively. The growing demand for quick, consistent, and accurate assessments highlights the need for an automatic essay evaluation system. Such a system can provide efficient grading, reduce subjectivity, and ensure standardized feedback, making it a valuable tool for both educational and professional settings.

DATASET

The dataset for this project was provided, and samples from the dataset are shown on the side.

#AUTHID	TEXT	cEXT	cNEU	cAGR	cCON	cOPN
1997_504	Well, right	n	y	y	n	y
1997_605	Well, here	n	n	y	n	n
1997_687	An open ke	n	y	n	y	y
1997_568	I can't beli	y	n	y	y	n
1997_688	Well, here	y	n	y	n	y
1997_722	Today. Ha	y	n	y	n	y
1997_724	Stream of	n	n	y	n	n
1997_724	The RTF30	n	n	n	y	y
1997_628	I'm really	y	y	n	y	y
1997_708	Today was	y	y	y	y	n
1997_665	Well, I am	y	y	y	y	y
1997_820	I have don	n	n	n	n	n
1997_780	well I am j	n	y	n	n	n
1997_606	Ok I've put	n	y	y	n	n
1997_606	sitting here	n	n	y	y	n
1997_111	always a p	y	n	y	n	n
1997_196	Psychologi	n	n	n	n	y
1997_636	1 Freestyl	n	n	y	n	y
1997_430	Well, I f	n	n	y	y	y
1997_475	Okay here	y	y	y	n	y
1997_356	I miss the	n	n	y	n	n
1997_530	I don't wa	n	y	y	n	y
1997_378	My neighb	n	y	y	y	n
1997_814	I'm feeling	y	n	y	y	n
1997_472	Wow, this	y	n	y	y	y
1997_339	As I sit her	y	n	y	y	n
1997_760	I just got o	y	n	y	y	n

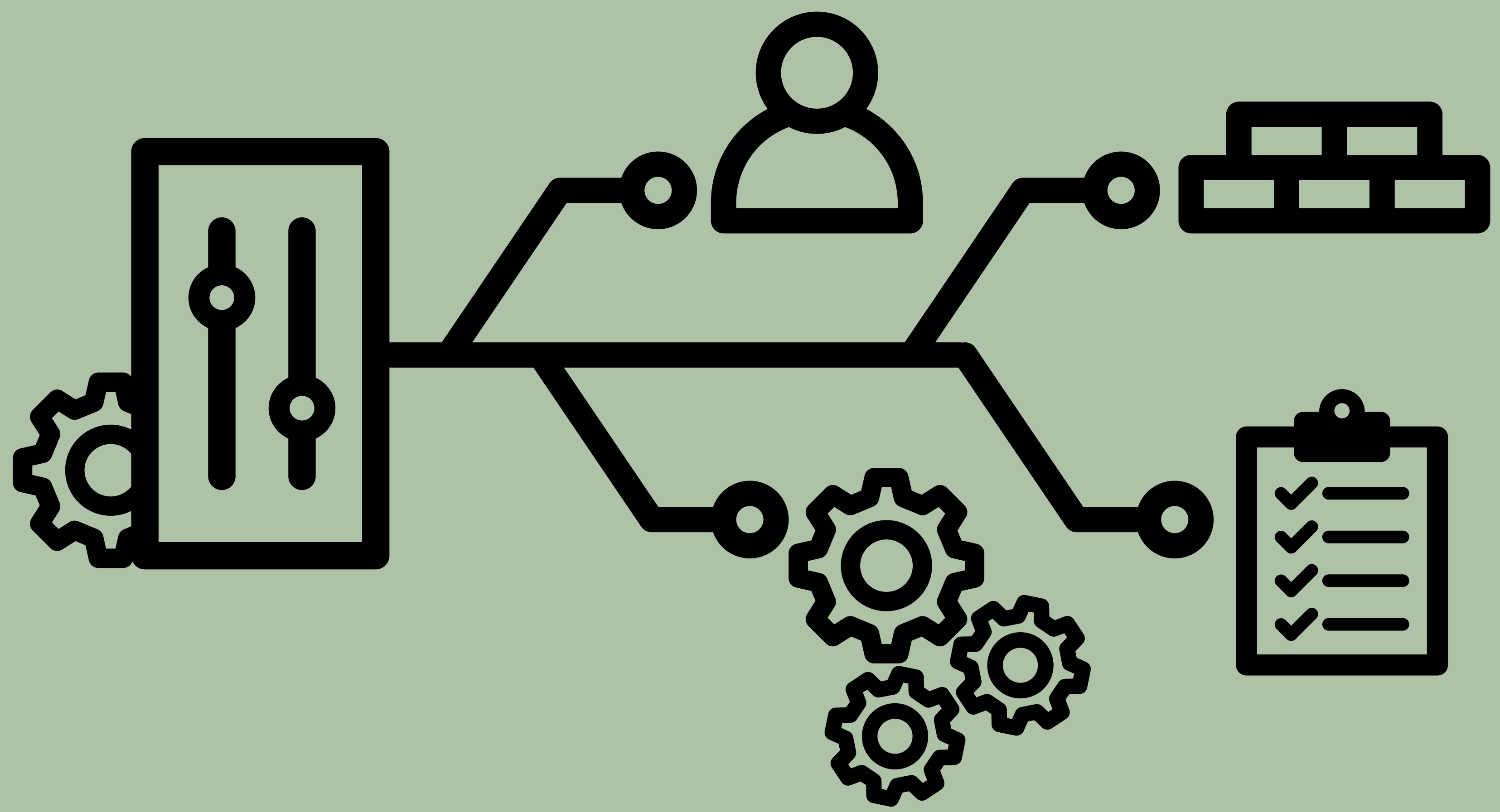
essays

Ready

METHODOLOGY

<u>1) Data Preprocessing:</u>

In the data preprocessing stage, irrelevant columns are removed, and relevant features, such as personality scores, are extracted. The text data is then prepared for analysis by normalizing and cleaning it. Outliers are identified and removed using statistical methods like the Interquartile Range (IQR) to enhance model performance. To address data imbalance, resampling techniques are applied, ensuring a balanced dataset. Finally, feature scaling is used to standardize the data, making it suitable for model training and preventing any feature from dominating due to differences in scale.



2) Model Training:

A suitable machine learning model (such as XGBRegressor) is trained using the processed features and labels (e.g., essay scores). The model learns patterns in the data to predict the desired outcome.

3) Text Evaluation Metrics:

Various metrics are calculated to evaluate the quality of the essay, including:

- Grammar Score: Analyzes grammatical correctness based on predefined error checks.
- Structure Score: Assesses the organization and clarity of ideas in the essay.
- Flow Score: Evaluates the smoothness of transitions between sentences and paragraphs.
- Word Count Score: Ensures the essay meets the required word count.
- Vocabulary Score: Measures the richness and diversity of vocabulary used in the essay.

4) Essay Correction and Model Inference:

The essay undergoes preprocessing to clean and correct the text, eliminating grammatical errors, spelling mistakes, and irrelevant characters. After preprocessing, the text is passed through a pre-trained model, such as BERT, to generate numerical embeddings. These embeddings are then used by a trained model to predict a score, providing an evaluation of the essay's quality.

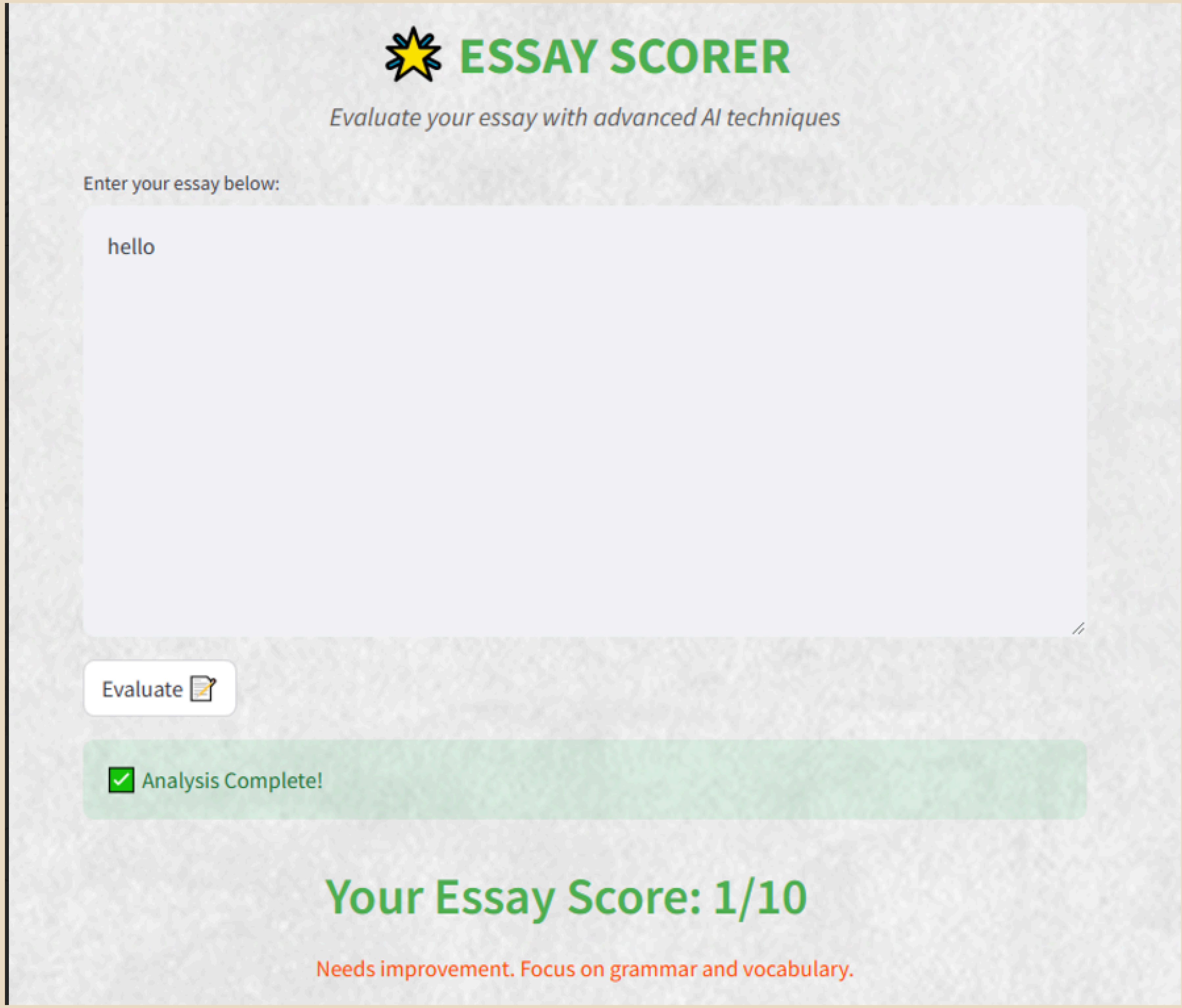
5) Final Score Calculation:

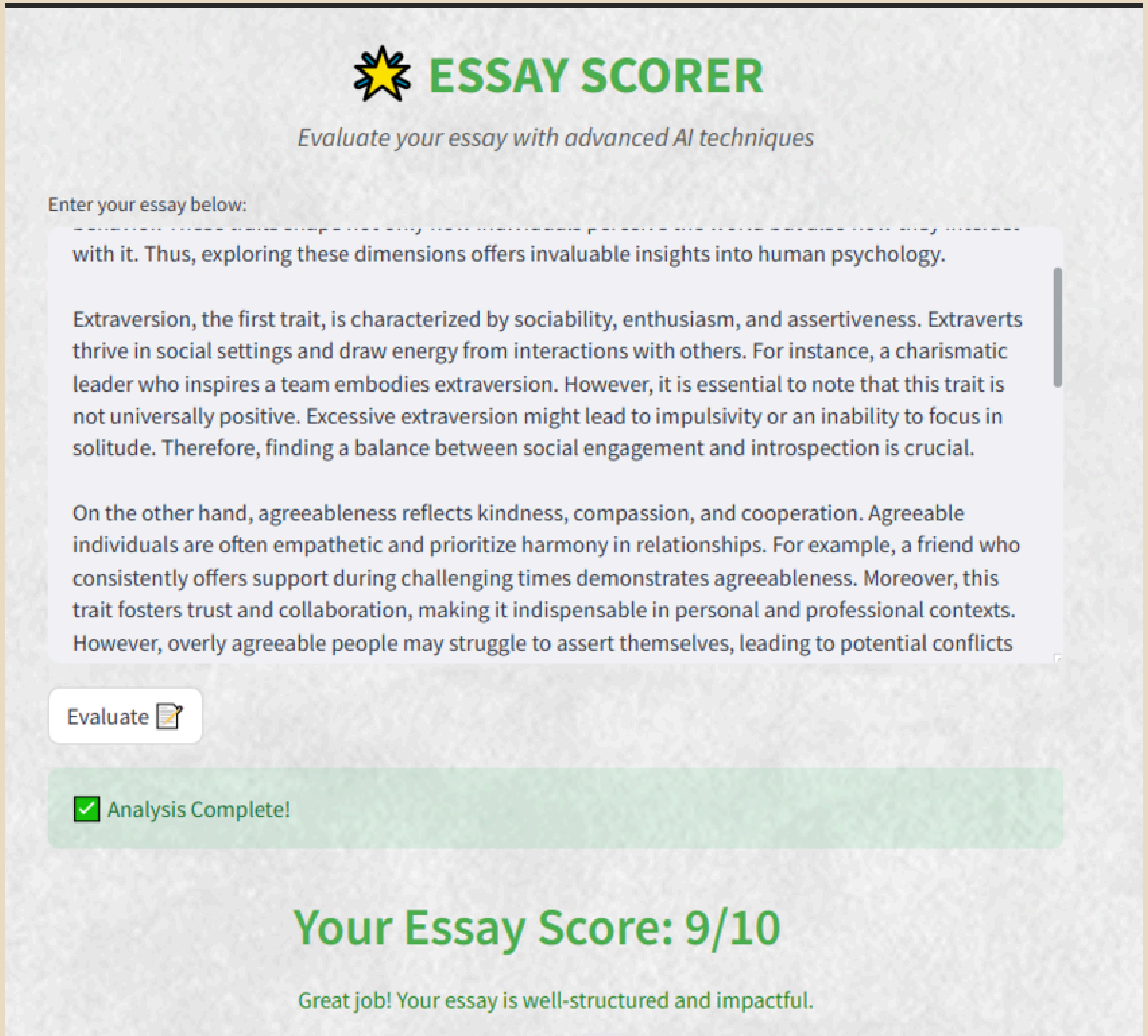
The final essay score is computed by combining the individual evaluation scores (grammar, structure, flow, word count, vocabulary) with the prediction from the mode

6) Streamlit Integration:

The entire essay evaluation process is integrated into a web application using Streamlit. Users can input their essays, trigger the evaluation process, and view the individual scores and final evaluation result through an interactive interface.

RESULTS


ESSAY SCORER
Evaluate your essay with advanced AI techniques
Enter your essay below:
hello
Evaluate
Analysis Complete!
Your Essay Score: 1/10
Needs improvement. Focus on grammar and vocabulary.


ESSAY SCORER
Evaluate your essay with advanced AI techniques
Enter your essay below:
with it. Thus, exploring these dimensions offers invaluable insights into human psychology.
Extraversion, the first trait, is characterized by sociability, enthusiasm, and assertiveness. Extraverts thrive in social settings and draw energy from interactions with others. For instance, a charismatic leader who inspires a team embodies extraversion. However, it is essential to note that this trait is not universally positive. Excessive extraversion might lead to impulsivity or an inability to focus in solitude. Therefore, finding a balance between social engagement and introspection is crucial.
On the other hand, agreeableness reflects kindness, compassion, and cooperation. Agreeable individuals are often empathetic and prioritize harmony in relationships. For example, a friend who consistently offers support during challenging times demonstrates agreeableness. Moreover, this trait fosters trust and collaboration, making it indispensable in personal and professional contexts. However, overly agreeable people may struggle to assert themselves, leading to potential conflicts
Evaluate
Analysis Complete!
Your Essay Score: 9/10
Great job! Your essay is well-structured and impactful.

USAGE

This project can be used in various domains where automated evaluation of textual content is needed. It can serve as a tool for educational institutions to assess essays, assignments, and written reports efficiently. Beyond education, it can be applied in content creation, recruitment, and customer support to evaluate written communication, ensuring quality, coherence, and correctness. The project helps automate tedious manual evaluation tasks, providing quick and consistent feedback.

FUTURE WORK

1) Improved Models: Fine-tuning or using advanced models like deep learning for better prediction accuracy.

2) More Evaluation Metrics: Adding semantic analysis or coherence scoring for a deeper essay assessment.

3) Multilingual Support: Extending the system to support multiple languages.
Better User Interface: Enhancing Streamlit UI with visualizations and detailed feedback.

4) Real-Time Feedback: Providing real-time corrections and suggestions while writing.

5) Platform Integration: Integrating with learning management systems like Moodle for automatic grading and feedback

CONCLUSION

This project successfully demonstrates the process of automating essay evaluation using machine learning and natural language processing techniques. By preprocessing the text, correcting errors, and generating numerical embeddings with a pre-trained model like BERT, the system efficiently evaluates essays based on various metrics such as grammar, structure, flow, vocabulary, and overall coherence. The integration of feature scaling, outlier removal, and resampling ensures that the model is robust and performs well across different data conditions. Ultimately, this approach streamlines the essay grading process, offering a scalable and objective method for academic assessments.

TEAM WORK:

Huzaifa: Data Preprocessing & Model Training
Responsible for loading and preprocessing the dataset, handling outliers, data resampling, and scaling features. Also trained and fine-tuned the machine learning model used for essay evaluation.

Zoha: Text Processing & Evaluation Metrics
Developed and implemented text correction functions, including grammar and spelling correction. Designed and calculated various essay evaluation metrics like grammar, structure, flow, and vocabulary.

Fatima: Model Inference & Streamlit UI
Managed the integration of the pre-trained model for generating essay embeddings and predictions. Designed and implemented the Streamlit UI, enabling users to interact with the essay evaluation system.

PPT Preparation
The presentation was collaboratively created by the team, with each member contributing to explaining their respective areas of work. The slides highlight key sections, such as the methodology, model performance, and future work, ensuring a comprehensive and cohesive presentation of the project.

Thank You